

B.Tech. Degree I Semester Examination November 2019

CS/EC/IT 19-200-0101 B CALCULUS
(2019 Scheme)

Maximum Marks : 60

Time : 3 Hours

PART A
(Answer ALL questions)

(8 × 3 = 24)

- I. (a) Solve $\cos y \frac{dy}{dx} + \tan x \sin y = \cos x$.
- (b) Solve $\frac{d^3 y}{dx^3} - \frac{d^2 y}{dx^2} + 100 \frac{dy}{dx} - 100y = 0$.
- (c) If $u = x^3 + y^3 - 3axy$, show that $y \frac{\partial^2 u}{\partial x^2} = x \frac{\partial^2 u}{\partial y^2}$.
- (d) If $u = x(1-y)$ and $v = xy$ prove that $JJ' = 1$ where $J = \frac{\partial(u,v)}{\partial(x,y)}$ and $J' = \frac{\partial(x,u)}{\partial(u,v)}$.
- (e) Change the order of integration in $\int_0^1 \int_{x^2}^x (2x^2 + y) dy dx$ and hence evaluate it.
- (f) Find the volume of the solid generated by revolving one arch of the cycloid $x = a(\theta + \sin \theta)$, $y = a(1 - \cos \theta)$
- (g) If $r = |\vec{r}|$ where $\vec{r} = xi + yj + zk$, prove that
(i) $\nabla r^n = n r^{n-2} \vec{r}$
(ii) $\nabla \cdot \vec{r} = 3$.
- (h) Determine the constant 'a' so that the vector $\vec{f} = (x+3y)i + (4-2z)j + (x+az)k$ is solenoidal.

PART B

(4 × 12 = 48)

- II. (a) Solve $(1+4xy+2y^2)dx + (3+4xy+2x^2)dy = 0$ so that $y(0) = 2$.
- (b) Solve $(D^2 + 4D + 5)y = e^{2x} + \cos 4x$.
OR
- III. (a) Solve $\frac{dx}{dt} + y = \sin t$, $\frac{dy}{dt} + x = \cos t$.
- (b) Solve $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = \log x + x$.
- IV. (a) If u is a homogeneous function of degree in x and y , show that $x^2 \frac{\partial^3 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = n(n-1)u$.
- (b) Find the dimensions of the rectangular box open at the top of maximum capacity whose surface area is 432 sq.cm.
OR
- V. (a) If $u = \log(\tan x + \tan y + \tan z)$ show that $\sin 2x \frac{\partial u}{\partial x} + \cos 2x \frac{\partial u}{\partial y} + \sin 2z \frac{\partial u}{\partial z} = 2$.

(P.T.O.)

- (b) The focal length of a mirror is given by the formula $\frac{1}{v} - \frac{1}{u} = \frac{2}{f}$. If equal small errors K are made in determination of u and v show that relative error in focal length f is given by $K\left(\frac{1}{v} + \frac{1}{u}\right)$.

- VI. (a) Find the perimeter of the cardioid $r = a(1 - \cos \theta)$.
 (b) Find by triple integration volume of the sphere $x^2 + y^2 + z^2 = a^2$.

OR

- VII. (a) Find the area of the surface generated by revolving about axis of x , the arc of the parabola $y^2 = 4ax$ from origin to the point $x = a$.
 (b) Evaluate the double integral $\iint xy(x+y)dx dy$ over the region bounded by $y = x$ and $y = x^2$.

- VIII. (a) Show that $\vec{F} = (6xy + z^3)\mathbf{i} + (3x^2 - z)\mathbf{j} + (3xz^2 - y)\mathbf{k}$ is irrotational. Also find its scalar potential.
 (b) If $\vec{A} = 2xy\mathbf{i} + 4z^2\mathbf{j} + xz\mathbf{k}$ and s is the rectangular parallelepiped bounded by $x = 0, y = 0, z = 0, x = 2, y = 1, z = 3$ use Gauss divergence theorem to evaluate $\iiint_s \vec{A} \cdot \hat{n} ds$.

OR

- IX. (a) Find the total workdone in moving a particle in a force field given by $\vec{F} = 3xy\mathbf{i} - 5z\mathbf{j} + 10x\mathbf{k}$ along the curve $x = t^2 + 1, y = 2t^2, z = t^3$ from $t = 1$ to $t = 2$.
 (b) State Stoke's theorem and use it to evaluate $\iint_s (\nabla \times \vec{F}) \cdot \hat{n} ds$ where $\vec{F} = (2x - y)\mathbf{i} - yz^2\mathbf{j} - y^2z\mathbf{k}$ and S is the upper half of the unit sphere $x^2 + y^2 + z^2 = 1$ and c its boundary.
